

SAMEER KHAN

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Senior Machine Learning Engineer specializing in AI, Deep Learning, and MLOps — delivering production-grade models, scalable pipelines, and high-impact AI solutions.

PROFILE SUMMARY

- With a **Bachelor's degree in Computer Science** and close to **8 years** of professional **experience**, I have a strong background in **machine learning, deep learning, NLP, Computer Vision**, and scalable **AI system**
- Proven expertise in designing and **deploying end-to-end AI pipelines**, from data ingestion and preprocessing to **model training**, deployment, and monitoring in production.
- Specialized in **Document AI & NLP** applications including document **classification**, **OCR-Free** extraction, **named entity recognition**, and **multimodal learning** using **CNN, BiLSTM, Transformers, Donut LLM**.
- Applied **deep learning architectures (CNNs, LSTMs, Transformers)** and embedding-based semantic search to deliver scalable document retrieval and recommendation systems.
- Hands-on experience with **MLOps & cloud platforms: AWS SageMaker, Databricks, Jenkins CI/CD, Docker, MLflow**, model packaging, and artifact governance.
- Built **vector search** pipelines and leveraging embeddings for intelligent document retrieval and RCA automation, reducing investigation time and improving accuracy.
- Proven ability to **deliver production-grade AI solutions** under tight deadlines, collaborating with cross-functional teams in fast-paced environments.

TECHNICAL SKILLS

LLM & NLP Tools : BERT, **Transformers**, NER, Hugging Face, Langchain

MLOps & Deployment: **SageMaker, Databricks, Jenkins**, Docker, MLflow, Terraform, CI/CD

Libraries & Platforms: Python, PyTorch, Tensorflow, Keras, Scikit-learn, FastAPI, **PySpark**, OpenCV

Databases & Cloud: MySQL, **DBFS**, AWS, Linux

WORK EXPERIENCE

Nov 2023 – Present: ICE Data Services, Hyderabad, India | Sr. Machine Learning Engineer

Key Result Areas

- Designed and deployed an **end-to-end multimodal ML pipeline** in **Databricks (PySpark)** for OCR-based document classification, from ingestion to production rollout.
- Built secure workflows to decrypt lender-provided images, extract text using **PyTesseract**, and merge with metadata and labels for **stratified train/val/test** datasets.
- Developed **preprocessing pipelines** with **GloVe embeddings**, **CNN-based image normalization**, and metadata enrichment to boost classification accuracy.
- Trained a **CNN + LSTM + Transformer hybrid ensemble** with **distributed Horovod training**, achieving **93% page-level** and **85% document-level** accuracy, while reducing training time by 30%.
- Integrated a **BERT Transformer** into the **ensemble model**, improving page-level accuracy from **93% to 94%**.
- Deployed **Donut LLM** as an **OCR-free alternative**, enabling direct text extraction without traditional OCR steps.
- Implemented an **embedding-based vector search pipeline** for root-cause analysis, replacing manual keyword search and cutting investigation time by 60%.
- Deployed production-ready models to **AWS SageMaker endpoints** using parameterized **Jenkins + Docker pipelines**, ensuring scalable delivery.

Aug 2021 – Nov 2023: Persistent Systems(Client: ICE Data Services), Hyderabad, India | Sr. ML Engineer

Key Result Areas

- Contributed to the **initial development** of an **OCR-based document classification** system project for financial documents, laying the foundation for production rollout at ICE.
- Designed data ingestion and **cleaning workflows** for scanned images, improving OCR text quality and consistency.
- Applied **OpenCV preprocessing** techniques (resize, thresholding, denoising, skew correction) to enhance accuracy of downstream models.
- Implemented OCR pipelines using Tesseract and AWS Textract for large-scale text extraction from scanned documents.
- Helped integrate **multimodal models** combining **CNN and LSTM** branches for document understanding.
- Assisted in implementing **BiLSTM encoder-decoder** components for **sequence vector** transformations.
- Developed **PySpark UDFs** for schema normalization, label encoding, and metadata creation.
- Supported **CI/CD** setup with AWS SageMaker for early-stage model deployment and monitoring.
- Gained early experience in **model packaging** and **versioning techniques**, working under senior guidance.

Aug 2019 – Mar 2021: Zyclyx, Hyderabad, India | Data Scientist
Client: Al Rajhi Bank (Onsite), Riyadh, Saudi Arabia | (Aug 2019 - Mar 2021)

Key Result Areas

- Gathered and analyzed **Arabic process-based financial documents** from the banks enterprise **data warehouse** for analytics and **model development**.
- **Designed and deployed** an **AI-powered document classification** and **information extraction** pipeline for **English-Arabic** financial documents.
- Fine-tuned a **ResNet50 image classification** model to categorize documents prior to downstream processing.
- Built an **OCR text extraction** pipeline using **Tesseract and TensorFlow object detection** to locate and label key fields on each page.
- Converted extracted fields into **JSON key-value pairs** and integrated them with APIs to validate against the bank's database records.
- Applied **Named Entity Recognition (NER)** with **GRU-based** models to identify customer, transaction, and compliance entities, improving accuracy of extraction.
- Designed and implemented a **Signature Verification System** using deep learning to detect fraudulent or mismatched signatures in offline documents.
- Deployed **ML models on-premise** using Flask, Gunicorn, and microservices, ensuring scalable inference for banking applications.

Feb 2017 – Aug 2019: VISSINDIA, Hyderabad, India | Python & Computer Vision Engineer

- Began career as a **Python developer**, gaining hands-on experience in scripting, data structures, and libraries such as NumPy, Pandas, and Matplotlib.
- Transitioned into **computer vision**, contributing to proof-of-concept (POC) projects for identity verification and object detection.
- Developed **face detection prototypes** using Haar Cascade (OpenCV), HOG (DLlib), and CNN-based methods.
- Built image **preprocessing pipelines** with morphological operations to reduce noise and improve OCR accuracy.
- Implemented **feature detection and matching algorithms** (SIFT, SURF, FLANN) for object similarity and tracking.
- Prototyped **TensorFlow object detection models** for multi-object tracking in images and videos.

PROJECTS

Vector Search RCA Tool (ICE Data Services) - Designed and deployed an embedding-based semantic search pipeline using production ensemble model vectors, replacing manual keyword search and reducing root-cause analysis time by 60%.

BERT & Donut LLM - OCR and OCR-Free Models (ICE Data Services) - Developed a Donut LLM-based extraction system as an OCR-free alternative, enabling direct document text retrieval without traditional OCR and improving processing efficiency.

CERTIFICATIONS & TRAINING

Advanced Deep Learning Specialization – Coursera (CNNs, RNNs, LSTMs, Sequence Models, Optimization)
Machine Learning Operations (MLOps) Specialization - Coursera(CI/CD, deployment, monitoring, ML pipelines)

ACADEMIC DETAILS

- **B.Tech. in Computer Science Engineering** - TKR Engineering College, JNTU, Hyderabad | 2016